Sahaj Bhadja

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Summary

Full-stack engineer with 2+ years shipping **AI-powered EdTech products, educator workflows, search, payments, and data platforms** using TypeScript, React, Node.js, PostgreSQL, Python, and AI product development.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Architected a multi-tenant AI education platform with PostgreSQL, FastAPI, and row-level security, supporting 500+ users while reducing dashboard latency from 1.2 seconds to 350 milliseconds for educators at scale.
- Owned LLM and data workflows through production deployment, monitoring, and iteration, processing 5,000+ monthly jobs with reconciliation, failure isolation, and CI/CD releases while sustaining 99%+ reliability at scale.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Shipped a full-stack AI grading platform with React and Python, serving 150+ educators and 10,000+ daily requests while reducing feedback turnaround 60% across text and handwritten submissions.
- Built prompt chains for rubric generation, grading, feedback, and reasoning across OpenAI, Anthropic, and Qwen models, adding concurrent processing, fallbacks, cost tracking, and production monitoring for reliable workflows.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Drove requirements through deployment for cross-platform microservices, collaborating with Microsoft and GitHub partners while improving API response times 30%, pipeline efficiency 68%, and technical debt 20% at scale.
- Implemented automated testing and staged production workflows across Linux, Windows, and macOS, supporting product quality through Git-based Agile sprints, code reviews, nightly builds, and integration testing at scale.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed React, Node.js, and TypeScript features across eight modular CRM services for customer records, billing, leads, permissions, and notifications, synchronizing two million records with 33% fewer conflicts.
- Used analytics and monitoring insights to guide product improvements, reducing resource waste 42% and increasing lead conversions 15% while participating in sprint delivery, testing, and production deployments.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Built React and TypeScript experiences with Flask APIs for booking, pricing, eligibility, discounts, and payment workflows, supporting 50,000+ records with sub-200-millisecond responses and real-time Firebase updates at scale.
- Designed end-to-end architecture with 12+ integration points and presented tradeoffs to the client, streamlining delivery 22% while automating refunds, disputes, and payment-status updates through idempotent webhooks at scale.

Skills

Full Stack: TypeScript, JavaScript, React, Next.js, Node.js, Express.js, Python, FastAPI, Flask
Data: PostgreSQL, SQL Server, MySQL, MongoDB, Redis, Firebase, SQL, data optimization
AI: LLM systems, prompt engineering, OpenAI, Claude, Qwen, PyTorch, OCR, semantic search
Product: EdTech, educator workflows, search, personalization, payments, analytics, monitoring, tooling
Engineering: Git, Agile/Scrum, CI/CD, code review, AWS, Azure, Docker, Kubernetes, REST APIs

Projects

Big-Brain Live | *AI Search, Real-Time Collaboration*

- Built a secure LLaMA-2 document assistant over 50,000+ records with semantic search, conversational question answering, and real-time collaboration, reducing analysis time 60% through low-latency synchronized workflows at scale.

FinNexus Live | *TypeScript, Payments, Data*

- Developed a Next.js and TypeScript payment platform supporting 50,000+ users and one million transactions, delivering real-time updates, secure Plaid integration, advanced filtering, scalable MongoDB, and Kubernetes deployment.

PiXMod Live | *Smart UI, AI Assistance*

- Shipped an AI-assisted image product using Next.js, CLIP, SAM, and Stable Diffusion, processing 1,000+ assets through responsive workflows, semantic discovery, scalable storage, and Stripe subscriptions for users.